



Gmail might be getting scheduled emails

A teardown of the latest Gmail APK has yielded references to scheduled emails feature which has been missing so far. <https://digit.in/apr19-83>



Google Photos Express Backup

Express Backup allows Android users to backup photos faster, in low-res for poor and infrequent net connectivity. <https://digit.in/apr19-84>

Intel Core i9-9980XE Extreme Edition

Can 18-cores take the crown?

The Intel Core i9-9980XE is an odd launch. Intel had been so far in the lead with

HEDT processors that their release cycles for these were about three years apart. X79 came out in 2011, followed by X99 in 2014 and X299 was unveiled at Computex 2017. It was expected that the successor would come somewhere in 2020 or in the latter part of 2019. AMD's Ryzen CPUs and their HEDT Threadripper CPUs threw a spanner in the works for Intel and out came another hurried launch. We say another because the X299 launch was also brought forward by a few months. The 18-core Intel Core i9-7980XE served its purpose and helped Intel maintain their lead in the HEDT segment since AMD's 16-core 1950X

would fall short in single and multi-threaded applications. However, the 7980XE had a \$1,999 price tag whereas the 1950X offered competitive performance for \$999 and it didn't reduce PCIe lanes either, so value-wise there was no beating AMD's Threadripper. A year down the line in 2018, AMD straight up doubled the core count to 32 with the 2990WX and Intel had no competitive solution in place. Why not the 9980XE? Well, it's the same chip as the 7980XE but with a few embellishments. A side-by-side comparison on the Intel ARK website shows that except for the frequency and the resulting TJunction and TDP values, nothing has changed. Does that mean that the Intel Core i9-9980XE is a bad CPU? Of course not. Here's why.

Rendering Performance

In our Blender benchmark, the core-counts matter the most so we see the Threadripper 2990WX finishing the run in the fastest time, followed by the 2970WX and then we have the



PERFORMANCE.....84
VALUE FOR MONEY.....75

Intel Core i9-9980XE in third spot. The 7980XE follows it by about 6 seconds. The difference between the no.1 spot and the 9980XE is 42 seconds. That's a lot for a short render benchmark. When the scene complexity increases, you'll

start seeing the time-to-completion gap widen much more making the 2990WX a better choice. In LuxMark, this gap is narrowed down to less than 1% because the render engine has been severely impacted by the Spectre and Meltdown mitigation patches. Moving to POV-Ray, we once again see similar results as observed in the Blender benchmark which pegs the Threadripper 2990WX in the top spot with the 9980XE coming in at 3rd.

While these results end up being disappointing, Intel has the not so secret weapon in the form of AVX512. PhotoWorxx is a CPU Hash Integer benchmark that makes use of AVX512 in all compatible CPUs and it really propels the Core i9-9980XE ahead of every other CPU. Even older gen Intel HEDT CPUs leapfrog over the Threadripper 2990WX when using AVX512.

Gaming Performance

You're much better off with a mainstream CPU if gaming is going to be your primary use case. We find that

the newer gen 9900K practically wipes the floor with all other CPUs in this scenario. In Far Cry 5 and Deus Ex: Mankind Divided, the lead is significant because these games tend to scale better with the CPU.

Price
₹1,84,000

Power consumption

Owing to the increased frequencies, the Intel Core i9-9980XE consumes about 10-11 Watts more than the Core i9-7980 at its peak. Which is quite low compared to the 2990WX which ends up consuming about 246 Watts. For desktop PCs, that isn't all that important but in a 2S server, this can end up being quite expensive in the long run. As long as the 12V rails on the power supply have the headroom, you should be able to run either CPU without issues.

Conclusion

Intel still retains the single-threaded crown but fails to retake its position in the multi-threaded arena. The newer Xeon W-3175X is Intel's attempt to regain its position but we haven't seen it in India yet. The 9980XE costs Rs.1,84,000 whereas the Threadripper 2990WX costs Rs.1,54,000. In most of our rendering and multi-core benchmarks the cheaper 2990WX scores more whereas in gaming and rare scenarios that would leverage Intel's proprietary instruction sets, the 9980XE would excel. Given that we're looking at the same chip being re-released, it simply doesn't seem to be worth the price. The CPU is very good but only if it had been priced lower than the competing AMD product.

—Mithun Mohandas

SPECIFICATIONS

CLOCK SPEED: 3.0 GHz | TURBO FREQUENCY: 4.4 GHz | CORES/THREADS: 18/36 | LITHOGRAPHY: 14nm | TDP: 165W | PCIe LANES: 44 | PCIe VERSION: 3.0 | MEMORY SUPPORT: DDR4-2666.

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